Cloud Computing (CS 308 1)

Reflection notes

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(1)

What I learned:

↳ Through this project, I learned how knowledge distillation works. A large teacher model (Phi-3.5, 3.8B) labels commands, and small student model (Pythia-70M) learns only from those teacher replies. This is sequence-level distillation only, because ollama give text only, no logits, so logit based distillation was not possible. The student reached 32% full frame accuracy, more than the teacher's 27% with 98% JSON validity, despite being about 54 times smaller.

= I also got hands-on practice with Kubernetes, docker, K3d, and SeaweedFS. I learned how to manage object storage, monitor pods, use health & readiness checks, scale with HPA, and do rolling updates & rollbacks.

(2) ## challenges I faced

↳ one challenge was disk-space. my windows C: drive fill to 20mb free. WSL & K3d storage was corrupted.

- Another challenge was spud. on 1 CPU. /Parse hung for over 120 seconds at 100% CPU.

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The student model also kept repeating JSON objects, because it was trained without an EOS token after the JSON.

↳ The load test also failed. With 4 users sending requests non-stop, Health probes timed out, Pods restarted, and most requests returned 499 and 502 errors.

(3) How I solved them

↳ For the disk problem, I move 5 large files from the C: drive to D: drive, to free space from 9.4 GB, and I recreated the k3d cluster clean.

↳ For the speed problem, I set torch.set_num_threads(1) in the service. After that, a single /parse took 0.5 seconds, and 50 sequential requests passed 50/50 in 23.7 seconds.

↳ For the Docker build, I added the PyTorch CPU index URL for torch, and a .dockerignore file. The build context fell from 2.075 GB to 329 KB, and image was 1.82 GB, under the 1 GB limit.

↳ The load-test problem is not fully solved model generation still blocks the FastAPI event loop so under heavy load the probes time out. I have written this as an open limitation in the report, not as a pass.

(4) Key Findings

Teacher (phi-3.5): JSON validity 93%, Intent & full frame 27%, student v1 (Pythia-70M) JSON validity 98%, intent 63%, date 73%.

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time 92%, place 78%, full from 32%
on trained base model : 0%
Labelling — 3000 commands labelled on all scans.
after cleaning, first 1500 used for training
Training V1 took 269.6 seconds, loss fell from
2.61 to 0.84.

Local test: 50/50 request passed, 0.47 seconds
per request, 303.6 MiB memory.

Local test: only 128 of 39,463 requests succeeded
Cost was 7.9 per hour for 2 replicas, which
is RS 2.06 per 1000 successful requests.

5 Conclusion

→ overall, this project gave me real hands on
experience in ML, cloud infrastructure & automate
I fact disk, speed & load problems, & fixing
them — & honestly reporting the one I could
not fix — helped me understanding how real
systems are build & tested.

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